

Case Studies

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[Practical]

Task 1: Create and Manage Files

Objective: Practice creating, listing, viewing, and deleting files

```
harshika@Harshika:~$ mkdir linux_task1
harshika@Harshika:~$ cd linux_task1
harshika@Harshika:~/linux_task1$ touch file1.txt file2.txt file3.txt file4.txt file5.txt
harshika@Harshika:~/linux_task1$ ls
file1.txt file2.txt file3.txt file4.txt file5.txt
harshika@Harshika:~/linux_task1$ ls -l
total 0
-rw-r--r-- 1 harshika harshika 0 Aug  6 16:40 file1.txt
-rw-r--r-- 1 harshika harshika 0 Aug  6 16:40 file2.txt
-rw-r--r-- 1 harshika harshika 0 Aug  6 16:40 file3.txt
-rw-r--r-- 1 harshika harshika 0 Aug  6 16:40 file4.txt
-rw-r--r-- 1 harshika harshika 0 Aug  6 16:40 file5.txt
harshika@Harshika:~/linux_task1$ nano file1.txt
harshika@Harshika:~/linux_task1$ cat file1.txt
This is my first Linux file.
harshika@Harshika:~/linux_task1$ rm file5.txt
harshika@Harshika:~/linux_task1$ ls
file1.txt file2.txt file3.txt file4.txt
harshika@Harshika:~/linux_task1$ |
```

Task 2: Copy and Move Files

Objective: Practice copying and moving files and directories

```

harshika@Harshika:~$ mkdir linux_task2
harshika@Harshika:~$ cd linux_task2
harshika@Harshika:~/linux_task2$ touch source.txt data.txt notes.txt
harshika@Harshika:~/linux_task2$ nano source.txt
harshika@Harshika:~/linux_task2$ mkdir backup
harshika@Harshika:~/linux_task2$ cp source.txt backup
harshika@Harshika:~/linux_task2$ ls backup
source.txt
harshika@Harshika:~/linux_task2$ cp data.txt backup/data_backup.txt
harshika@Harshika:~/linux_task2$ ls backup
data_backup.txt  source.txt
harshika@Harshika:~/linux_task2$ mv notes.txt backup/
harshika@Harshika:~/linux_task2$ ls
backup  data.txt  source.txt
harshika@Harshika:~/linux_task2$ ls backup
data_backup.txt  notes.txt  source.txt
harshika@Harshika:~/linux_task2$ mv source.txt original.txt
harshika@Harshika:~/linux_task2$ ls
backup  data.txt  original.txt
harshika@Harshika:~/linux_task2$ ls backup
data_backup.txt  notes.txt  source.txt
harshika@Harshika:~/linux_task2$ tree
.
├── backup
│   ├── data_backup.txt
│   ├── notes.txt
│   └── source.txt
├── data.txt
└── original.txt

2 directories, 5 files
harshika@Harshika:~/linux_task2$ |

```

Task 3: Working with head and tail

Objective: Practice viewing specific portions of a file.

```
harshika@Harshika:~$ mkdir linux_task3
harshika@Harshika:~$ cd linux_task3
harshika@Harshika:~/linux_task3$ touch students.txt
harshika@Harshika:~/linux_task3$ nano students.txt
harshika@Harshika:~/linux_task3$ cat students.txt
Ravinder
Srilatha
Varshika
Harshika
Santhvin
Chinni
Manu
Bhanu Priya
Nikshitha
Pranitha
Angela
Keerthi Priya
Jhansi Priya
Lahari
Sreedhar
harshika@Harshika:~/linux_task3$ head -5 students.txt
Ravinder
Srilatha
Varshika
Harshika
Santhvin
harshika@Harshika:~/linux_task3$ head students.txt
Ravinder
Srilatha
Varshika
Harshika
Santhvin
Chinni
Manu
Bhanu Priya
Nikshitha
Pranitha
harshika@Harshika:~/linux_task3$ tail -5 students.txt
Angela
Keerthi Priya
Jhansi Priya
```

```
Pranitha
Angela
Keerthi Priya
Jhansi Priya
Lahari
Sreedhar
harshika@Harshika:~/linux_task3$ head -5 students.txt
Ravinder
Srilatha
Varshika
Harshika
Santhvin
harshika@Harshika:~/linux_task3$ head students.txt
Ravinder
Srilatha
Varshika
Harshika
Santhvin
Chinni
Manu
Bhanu Priya
Nikshitha
Pranitha
harshika@Harshika:~/linux_task3$ tail -5 students.txt
Angela
Keerthi Priya
Jhansi Priya
Lahari
Sreedhar
harshika@Harshika:~/linux_task3$ tail students.txt
Chinni
Manu
Bhanu Priya
Nikshitha
Pranitha
Angela
Keerthi Priya
Jhansi Priya
Lahari
Sreedhar
harshika@Harshika:~/linux_task3$ |
```

Task 4: Organize Files

Objective: Organize files into different directories using mv

```
harshika@Harshika:~$ mkdir organization
harshika@Harshika:~$ cd organization
harshika@Harshika:~/organization$ touch file1.txt file2.txt file3.txt image1.jpg image2.jpg
harshika@Harshika:~/organization$ ls
file1.txt file2.txt file3.txt image1.jpg image2.jpg
harshika@Harshika:~/organization$ mkdir documents
harshika@Harshika:~/organization$ mkdir images
harshika@Harshika:~/organization$ mv *.txt documents/
harshika@Harshika:~/organization$ ls documents
file1.txt file2.txt file3.txt
harshika@Harshika:~/organization$ mv *.jpg images/
harshika@Harshika:~/organization$ ls images
image1.jpg image2.jpg
harshika@Harshika:~/organization$ ls
documents images
harshika@Harshika:~/organization$ mv documents/file1.txt documents/notes.txt
harshika@Harshika:~/organization$ ls documents
file2.txt file3.txt notes.txt
harshika@Harshika:~/organization$ mv images/image1.jpg images/photo.jpg
harshika@Harshika:~/organization$ ls images
image2.jpg photo.jpg
harshika@Harshika:~/organization$ tree
.
├── documents
│   ├── file2.txt
│   ├── file3.txt
│   └── notes.txt
└── images
    ├── image2.jpg
    └── photo.jpg

3 directories, 5 files
harshika@Harshika:~/organization$ |
```

Task 5: Directory Management

Objective: Practice creating nested directories and removing them

```

harshika@Harshika:~$ mkdir Project
harshika@Harshika:~$ cd Project
harshika@Harshika:~/Project$ mkdir source/
harshika@Harshika:~/Project$ mkdir source/c/
harshika@Harshika:~/Project$ mkdir source/python/
harshika@Harshika:~/Project$ mkdir docs/
harshika@Harshika:~/Project$ mkdir backup/
harshika@Harshika:~/Project$ touch source/c/file1.c
harshika@Harshika:~/Project$ touch source/c/file2.c
harshika@Harshika:~/Project$ touch source/c/file3.c
harshika@Harshika:~/Project$ ls source/c/
file1.c file2.c file3.c
harshika@Harshika:~/Project$ touch source/python/program1.py source/python/program2.py
harshika@Harshika:~/Project$ ls source/python/
program1.py program2.py
harshika@Harshika:~/Project$ tree
.
├── backup
├── docs
├── source
│   ├── c
│   │   ├── file1.c
│   │   ├── file2.c
│   │   └── file3.c
│   └── python
│       ├── program1.py
│       └── program2.py

```

```

6 directories, 5 files
harshika@Harshika:~/Project$ rmdir backup
harshika@Harshika:~/Project$ ls
docs source
harshika@Harshika:~/Project$ mkdir backup
harshika@Harshika:~/Project$ touch backup/data.txt
harshika@Harshika:~/Project$ rmdir backup
rmdir: failed to remove 'backup': Directory not empty
harshika@Harshika:~/Project$ rm backup/data.txt
harshika@Harshika:~/Project$ ls backup
harshika@Harshika:~/Project$ rmdir backup
harshika@Harshika:~/Project$ ls
docs source

```

```

harshika@Harshika:~/Project$ mkdir backup
harshika@Harshika:~/Project$ touch backup/data.txt
harshika@Harshika:~/Project$ rmdir backup
rmdir: failed to remove 'backup': Directory not empty
harshika@Harshika:~/Project$ rm backup/data.txt
harshika@Harshika:~/Project$ ls backup
harshika@Harshika:~/Project$ rmdir backup
harshika@Harshika:~/Project$ ls
docs source
harshika@Harshika:~/Project$ tree

```

```

.
├── docs
├── source
│   ├── c
│   │   ├── file1.c
│   │   ├── file2.c
│   │   └── file3.c
│   └── python
│       ├── program1.py
│       └── program2.py

```

```

5 directories, 5 files
harshika@Harshika:~/Project$ |

```

Task 6: File Permissions Using chmod

Objective: Understand Linux file permissions

```
harshika@Harshika:~$ mkdir linux_task4
harshika@Harshika:~$ cd linux_task4
harshika@Harshika:~/linux_task4$ touch public.txt private.txt script.sh
harshika@Harshika:~/linux_task4$ nano public.txt
harshika@Harshika:~/linux_task4$ nano private.txt
harshika@Harshika:~/linux_task4$ nano script.sh
harshika@Harshika:~/linux_task4$ ls -l
total 12
-rw-r--r-- 1 harshika harshika 24 Aug  6 17:45 private.txt
-rw-r--r-- 1 harshika harshika 23 Aug  6 17:45 public.txt
-rw-r--r-- 1 harshika harshika 31 Aug  6 17:46 script.sh
harshika@Harshika:~/linux_task4$ chmod 600 private.txt
harshika@Harshika:~/linux_task4$ chmod 644 public.txt
harshika@Harshika:~/linux_task4$ chmod +x script.sh
harshika@Harshika:~/linux_task4$ ls -l
total 12
-rw----- 1 harshika harshika 24 Aug  6 17:45 private.txt
-rw-r--r-- 1 harshika harshika 23 Aug  6 17:45 public.txt
-rwxr-xr-x 1 harshika harshika 31 Aug  6 17:46 script.sh
harshika@Harshika:~/linux_task4$ ./script.sh
Hello Linux
harshika@Harshika:~/linux_task4$ ls -l
total 12
-rw----- 1 harshika harshika 24 Aug  6 17:45 private.txt
-rw-r--r-- 1 harshika harshika 23 Aug  6 17:45 public.txt
-rwxr-xr-x 1 harshika harshika 31 Aug  6 17:46 script.sh
harshika@Harshika:~/linux_task4$ chmod -x script.sh
harshika@Harshika:~/linux_task4$ ls -l
total 12
-rw----- 1 harshika harshika 24 Aug  6 17:45 private.txt
-rw-r--r-- 1 harshika harshika 23 Aug  6 17:45 public.txt
-rw-r--r-- 1 harshika harshika 31 Aug  6 17:46 script.sh
harshika@Harshika:~/linux_task4$ |
```

Task 7: File Permission Challenge

Objective: Understand how permissions affect files and directories

```
harshika@Harshika:~$ mkdir permission_lab
harshika@Harshika:~$ cd permission_lab
harshika@Harshika:~/permission_lab$ touch test.txt
harshika@Harshika:~/permission_lab$ nano test.txt
harshika@Harshika:~/permission_lab$ ls -l
total 4
-rw-r--r-- 1 harshika harshika 21 Aug  6 17:51 test.txt
harshika@Harshika:~/permission_lab$ chmod u-r test.txt
harshika@Harshika:~/permission_lab$ ls -l
total 4
--w-r--r-- 1 harshika harshika 21 Aug  6 17:51 test.txt
harshika@Harshika:~/permission_lab$ chmod u+r test.txt
harshika@Harshika:~/permission_lab$ ls -l
total 4
-rw-r--r-- 1 harshika harshika 21 Aug  6 17:51 test.txt
harshika@Harshika:~/permission_lab$ chmod u-w test.txt
harshika@Harshika:~/permission_lab$ ls -l
total 4
-r--r--r-- 1 harshika harshika 21 Aug  6 17:51 test.txt
harshika@Harshika:~/permission_lab$ nano test.txt
harshika@Harshika:~/permission_lab$ chmod u+w test.txt
harshika@Harshika:~/permission_lab$ ls -l
total 4
-rw-r--r-- 1 harshika harshika 21 Aug  6 17:51 test.txt
harshika@Harshika:~/permission_lab$ ./test.txt
-bash: ./test.txt: Permission denied
harshika@Harshika:~/permission_lab$ |
```

Task 8: Backup and Restore

Objective: Create backups and restore files using cp

```

harshika@Harshika:~$ mkdir linux_task6
harshika@Harshika:~$ cd linux_task6
harshika@Harshika:~/linux_task6$ touch important.txt
harshika@Harshika:~/linux_task6$ nano important.txt
harshika@Harshika:~/linux_task6$ mkdir backup
harshika@Harshika:~/linux_task6$ cp important.txt backup/
harshika@Harshika:~/linux_task6$ tree
.
├── backup
│   └── important.txt
└── important.txt

2 directories, 2 files
harshika@Harshika:~/linux_task6$ mv backup/important.txt backup/important_backup.txt
harshika@Harshika:~/linux_task6$ tree
.
├── backup
│   └── important_backup.txt
└── important.txt

2 directories, 2 files
harshika@Harshika:~/linux_task6$ rm important.txt
harshika@Harshika:~/linux_task6$ tree
.
├── backup
│   └── important_backup.txt
└──

2 directories, 1 file
harshika@Harshika:~/linux_task6$ ls
backup
harshika@Harshika:~/linux_task6$ cp backup/important_backup.txt .
harshika@Harshika:~/linux_task6$ ls
backup  important_backup.txt
harshika@Harshika:~/linux_task6$ tree
.
├── backup
│   └── important_backup.txt
└── important_backup.txt

2 directories, 2 files
harshika@Harshika:~/linux_task6$ mv important_backup.txt important_restored.txt

```

```

harshika@Harshika:~/linux_task6$ mv important_backup.txt important_restored.txt
harshika@Harshika:~/linux_task6$ cat important_restored.txt
Operating System Lab
Linux Commands
Backup and Restore
Learning cp command
Learning mv command
Learning chmod command
Learning mkdir command
Learning rm command
Learning directories
End of File
harshika@Harshika:~/linux_task6$ |

```

Task 9: Complete Linux File Management Challenge

Objective: Use almost all the commands together.

```

harshika@Harshika:~$ mkdir linux_project
harshika@Harshika:~$ cd linux_project
harshika@Harshika:~/linux_project$ mkdir source backup docs
harshika@Harshika:~/linux_project$ touch source/main.c source/helper.c source/README.txt
harshika@Harshika:~/linux_project$ ls source/
README.txt  helper.c  main.c
harshika@Harshika:~/linux_project$ nano source/README.txt
harshika@Harshika:~/linux_project$ head -5 source/README.txt
Operating System Lab
Linux Commands
File Management
Directory Management
Permissions
harshika@Harshika:~/linux_project$ tail -5 source/README.txt
Backup Folder
Docs Folder
Learning Commands
Operating Systems
End of File
harshika@Harshika:~/linux_project$ tree
.
├── backup
├── docs
└── source
    ├── README.txt
    ├── helper.c
    └── main.c

4 directories, 3 files
harshika@Harshika:~/linux_project$ cp source/main.c backup/
harshika@Harshika:~/linux_project$ ls backup/
main.c
harshika@Harshika:~/linux_project$ ls
backup  docs  source
harshika@Harshika:~/linux_project$ cp source/README.txt docs/
harshika@Harshika:~/linux_project$ ls docs
README.txt
harshika@Harshika:~/linux_project$ mv source/helper.c source/functions.c
harshika@Harshika:~/linux_project$ ls source
README.txt  functions.c  main.c
harshika@Harshika:~/linux_project$ chmod 644 source/README.txt

```

```

harshika@Harshika:~$ mkdir linux_project
harshika@Harshika:~$ cd linux_project
harshika@Harshika:~/linux_project$ mkdir source backup docs
harshika@Harshika:~/linux_project$ touch source/main.c source/helper.c source/README.txt
harshika@Harshika:~/linux_project$ ls source/
README.txt  helper.c  main.c
harshika@Harshika:~/linux_project$ nano source/README.txt
harshika@Harshika:~/linux_project$ head -5 source/README.txt
Operating System Lab
Linux Commands
File Management
Directory Management
Permissions
harshika@Harshika:~/linux_project$ tail -5 source/README.txt
Backup Folder
Docs Folder
Learning Commands
Operating Systems
End of File
harshika@Harshika:~/linux_project$ tree
.
├── backup
├── docs
└── source
    ├── README.txt
    ├── helper.c
    └── main.c

4 directories, 3 files
harshika@Harshika:~/linux_project$ cp source/main.c backup/
harshika@Harshika:~/linux_project$ ls backup/
main.c
harshika@Harshika:~/linux_project$ ls
backup  docs  source
harshika@Harshika:~/linux_project$ cp source/README.txt docs/
harshika@Harshika:~/linux_project$ ls docs
README.txt
harshika@Harshika:~/linux_project$ mv source/helper.c source/functions.c
harshika@Harshika:~/linux_project$ ls source
README.txt  functions.c  main.c
harshika@Harshika:~/linux_project$ chmod 644 source/README.txt

```

Task 10: Mini Linux Administration Challenge

Objective: Complete a real-world-style file management exercise

```
harshika@Harshika:~/linux_project$ mkdir student_portal
harshika@Harshika:~/linux_project$ cd student_portal
harshika@Harshika:~/linux_project/student_portal$ mkdir src docs backup logs config
harshika@Harshika:~/linux_project/student_portal$ ls
backup  config  docs  logs  src
harshika@Harshika:~/linux_project/student_portal$ touch src/main.c src/student.c docs/README.txt config/config.txt logs/system.log
harshika@Harshika:~/linux_project/student_portal$ tree
.
├── backup
├── config
│   └── config.txt
├── docs
│   └── README.txt
├── logs
│   └── system.log
└── src
    ├── main.c
    └── student.c

6 directories, 5 files
harshika@Harshika:~/linux_project/student_portal$ nano logs/system.log
harshika@Harshika:~/linux_project/student_portal$ head -5 logs/system.log
System Started
User Login
Configuration Loaded
Database Connected
Service Started
harshika@Harshika:~/linux_project/student_portal$ tail -5 logs/system.log
Permissions Verified
Files Synced
System Healthy
System Shutdown

harshika@Harshika:~/linux_project/student_portal$ nano logs/system.log
harshika@Harshika:~/linux_project/student_portal$ tail -5 logs/system.log
Memory Checked
Permissions Verified
Files Synced
System Healthy
System Shutdown
harshika@Harshika:~/linux_project/student_portal$ cp docs/README.txt backup/
```

```
harshika@Harshika:~/linux_project/student_portal$ cp docs/README.txt backup/
harshika@Harshika:~/linux_project/student_portal$ ls backup
README.txt
harshika@Harshika:~/linux_project/student_portal$ mv backup/README.txt backup/README_backup.txt
harshika@Harshika:~/linux_project/student_portal$ ls backup
README_backup.txt
harshika@Harshika:~/linux_project/student_portal$ chmod 600 config/config.txt
harshika@Harshika:~/linux_project/student_portal$ ls -l config
total 0
-rw----- 1 harshika harshika 0 Aug  7 03:25 config.txt
harshika@Harshika:~/linux_project/student_portal$ chmod 644 docs/README.txt
harshika@Harshika:~/linux_project/student_portal$ ls -l docs
total 0
-rw-r--r-- 1 harshika harshika 0 Aug  7 03:25 README.txt
harshika@Harshika:~/linux_project/student_portal$ rm backup/README_backup.txt
harshika@Harshika:~/linux_project/student_portal$ ls backup
harshika@Harshika:~/linux_project/student_portal$ rmdir backup
harshika@Harshika:~/linux_project/student_portal$ ls
config  docs  logs  src
harshika@Harshika:~/linux_project/student_portal$ tree
.
├── config
│   └── config.txt
├── docs
│   └── README.txt
├── logs
│   └── system.log
└── src
    ├── main.c
    └── student.c

5 directories, 5 files
harshika@Harshika:~/linux_project/student_portal$ |
```